XIAMEN UNIVERSITY MALAYSIA

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I/We hereby confirm that all the references or sources of citations have been correctly listed or presented and I/we clearly understand the serious consequence caused by any intentional or unintentional misconduct.

This work is not made on any work of other students (past or present), and it has not been submitted to any other courses or institutions before.

Signature of the team members: *KHOO, ENG, NG, KOH, ZHANG, DUAN*

Date:26/03/2026

1.0 Introduction

ViTrox Corporation Berhad was founded in 2000 in Penang, Malaysia by Dato' Chu Jenn Weng and has been a leader in semiconductor and electronics inspection in Penang for 26 years. The business focuses on Automated Machine Vision Inspection Systems, combining robotics, IoT, and AI to promote Industry 5.0 autonomous manufacturing. The V510i Advanced 3D Automated Optical Inspection (AOI), their most impactful product, uses True 3D imaging and deep learning to identify complex defects. The award-winning V810i S3 DL, their most recent innovation, offers high-speed 3D X-Ray Inspection for hidden solder joints in large-scale manufacturing.

1.1 Problem Facing

In order to maintain high-accuracy vision algorithms for global clients, the AOI department must overcome a "Development-to-Deployment" bottleneck. Among the main problems are:

- Manual Verification: To avoid affecting detection capabilities, developers must manually test each code change against thousands of "Golden Sample" images.
- Communication Lag: When field engineers report "Issues" through GitHub, the internal team responds slowly due to a high volume of triaging.

1.2 Proposed Agentic Solution

Agentic AI is the shift from text-based, passive models to autonomous, useful systems. In contrast to standard AI, an agent can actively solve real-world engineering problems by using external tools, making independent decisions, and carrying out multi-step workflows to accomplish goals without human guidance. We suggest a three-part agentic workflow to enhance the AOI Department's software lifecycle:

1. **Automated Quality Assurance and Test Case Generation:** When new inspection code is committed to GitHub, the agent starts an evaluation, analyses changes, and creates test cases to check algorithm accuracy.
2. **Automated Team Notification System:** The agent monitors GitHub activity and sends email updates to the AOI engineering team.
3. **Autonomous Issue Resolution and Closing:** As the first responder to GitHub "Issues", the agent analyses previous data for solutions, conducts verification tests, and recommends closing the problem once it is solved.

1.3 Workflow

Step 1: **Intelligence Monitoring and Classification.** It starts with a GitHub Trigger node that tracks the repository activity. An IF Node classifies the event right away. If it finds a New Issue, the agent starts the bug-resolution process; If it finds a Pull Request, it uses different code review logic.

Step 2: **Validation and First Notification.** The second IF Node confirms whether the is a new issue. If True, the system triggers a Gmail node to send a notification to the lead engineer, then the checking process will begin.

Step 3: **Autonomous Debugging and Code Correction.** The agent collects technical error logs and relevant source code using the "Get an Issue" and "Get a File" nodes. The system analyses the bugs and automatically generates a corrected version of the algorithm code using the AI agent node (Ollama).

Step 4: **5 Test Cases for Automated Testing.** The agent goes into a loop to create five different test cases that specifically target the failure to make sure the fix is stable. Pytest or JavaScript nodes are used to confirm that the code is runnable and logically sound.

Step 5: **Aggregation and Resolution of Results**. The test pass or fail results are compiled by an Aggregate node. The reporter receives an email via Gmail with the final test summary and the updated code. The agent suggest closing the GitHub issue and finishing the autonomous cycle once it verifies the fix is effective.

1.4 Evaluation & Experiment

We had created three tests evaluate the AI Agent's ability to handle various programming patterns. In order to test the agent's ability to detect semantic errors, the Calculator experiment uses an international "silent bug" in which the division operator "/" is replaced by the modulo operator "%". The agent must map several if-elif thresholds and manage possible ValueError exceptions in the Grading experiment, which focuses on structured conditional logic and range validation. Lastly, the Weather experiment tests input sanitization and data structure handling, specifically requiring the agent to understand the need for ".strip()" and ".title()" techniques to preserve code robustness. When combined, these scripts offer a valid testbed for evaluating logic mapping, debugging, and defensive programming.

1.5 Result & Discussion

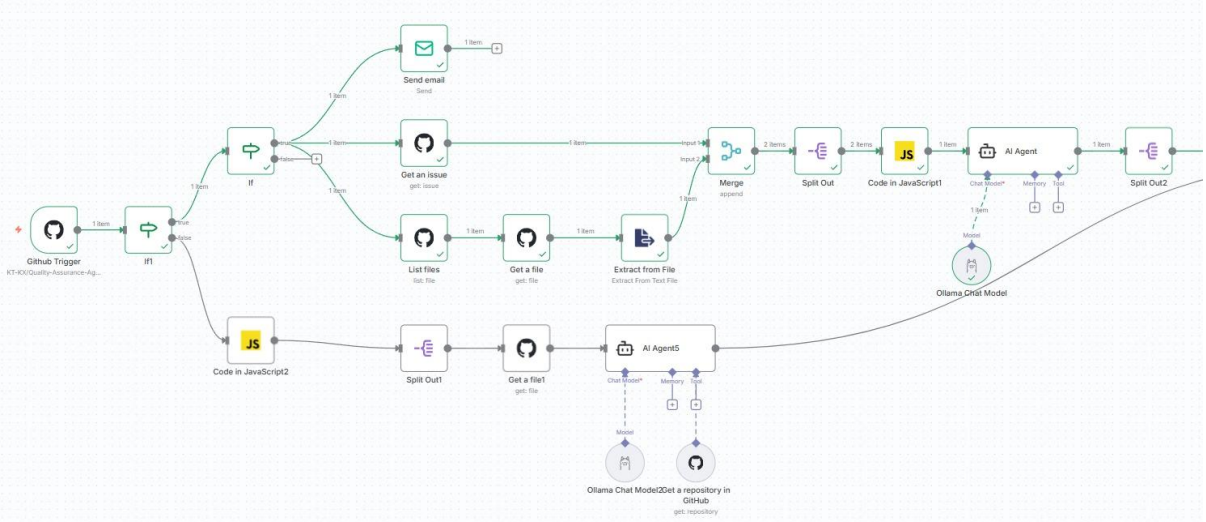
Our testing focuses primarily on "automated closed-loop verification" logic. An AI that mimics a Senior Architect gets code via GitHub, analyses it, and creates five Pytest cases for local execution. In order to assess logical accuracy in a matter of seconds, it automatically compares "expected output" with "actual results", bypassing manual presets.

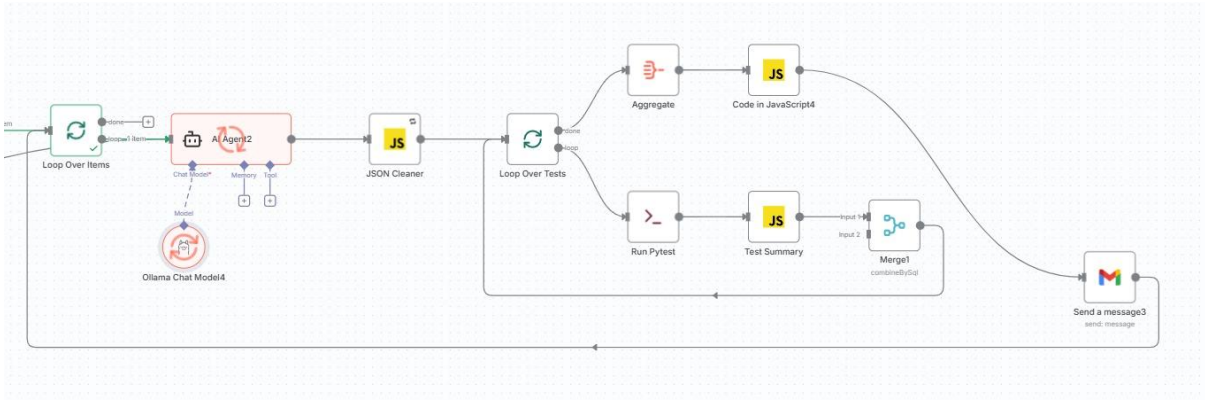
We designed three experiments to validate this. The modulo operator "%" was used in the Calculator experiment in place of division. Basic operations succeeded, but the AI correctly identified the logical error in the "division by zero" test, showing that it failed. This shows the system's capacity to find hidden flaws. Robustness and business logic were the main topics of the Weather and Grading experiments. The AI covered all conditional paths and passed robustness tests for irregular inputs, with all results marked PASSED.

In short, the test cases cover the dimensions of resilience, exception, boundary, and functionality. A FAIL shows an unhandled crash or logical blind spot, whereas a PASS verifies that the logic matches criteria. In the end, this "monitor-test-notify" cycle demonstrates efficacy through its quick feedback loop and guarantees correctness through AI-generated assertions.

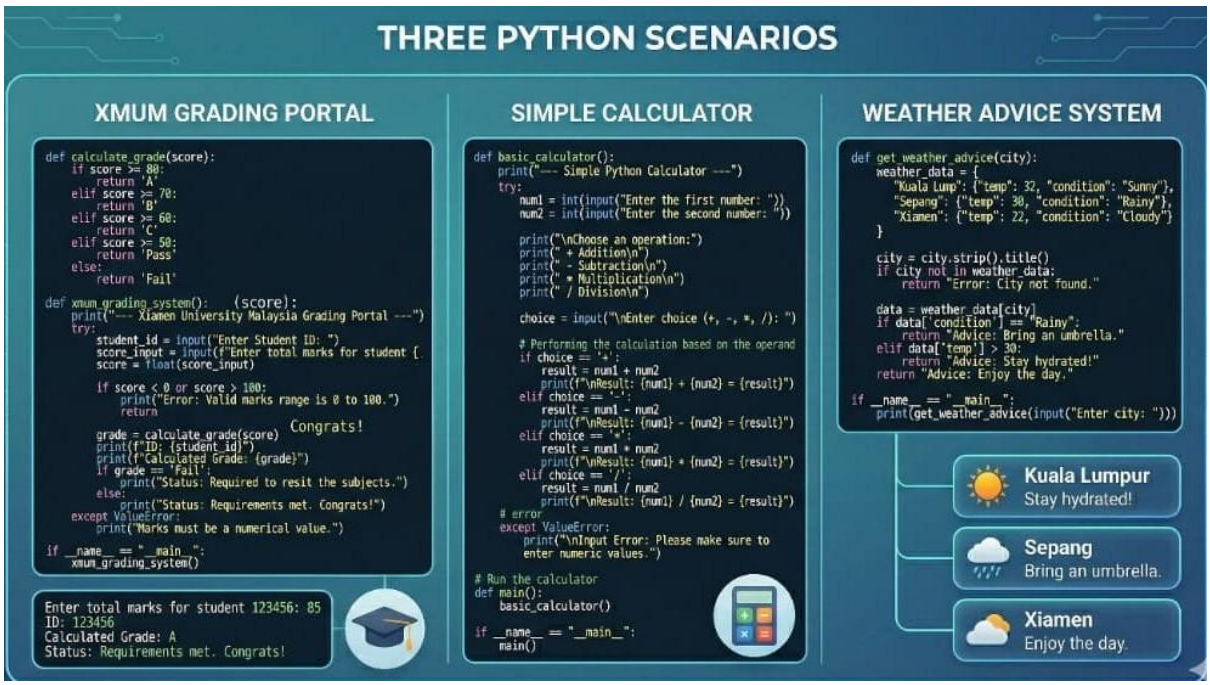
1.6 Appendix

Workflow





Experiment Test Case Code



Test Cases Results

QA Test Report

No	ID	Test Description	Status	Output
1	TC1	Verify that the function returns 'A' for a score of 80.	PASSED	Success
2	TC2	Verify that the function returns 'B' for a score of 75.	PASSED	Success
3	TC3	Verify that the function returns 'Pass' for a score of 50.	PASSED	Success
4	TC4	Verify that the function returns 'Fail' for a score of 40.	PASSED	Success
5	TC5	Verify that the function handles invalid input (non-numeric value).	PASSED	Success

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This email was sent automatically with n8n

QA Test Report

No	ID	Test Description	Status	Output
1	TC1	Verify the calculator performs addition correctly	PASSED	Success
2	TC2	Verify the calculator handles division by zero	FAILED	========== short test summary info ========== FAILED AppData/Local/Temp/test_d658d2af-89ee-4100-89c8-b4248c035b90.py::test_division_by_zero ========== 1 failed in 0.26s ==========
3	TC3	Verify the calculator handles non-numeric input gracefully	PASSED	Success
4	TC4	Verify the calculator performs subtraction correctly	PASSED	Success
5	TC5	Verify the calculator performs multiplication correctly	PASSED	Success

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QA Test Report

No	ID	Test Description	Status	Output
1	TC1	Verify the function returns the correct advice for a city with rainy weather.	PASSED	Success
2	TC2	Verify the function returns the correct advice for a city with temperatures over 30 degrees.	PASSED	Success
3	TC3	Verify the function returns the correct advice for a city with temperatures below or equal to 30 degrees.	PASSED	Success
4	TC4	Verify the function returns an error message for a city not found in the weather data.	PASSED	Success
5	TC5	Verify the function handles leading/trailing spaces in the city input.	PASSED	Success

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APPENDIX 1

MARKING RUBRICS

Component Title	Business Understanding and Proposal				Percentage (%)	20	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (10-9)	Good (8-7)	Average (6-5)	Need Improvement (4-3)	Poor (2-0)		
Agentic Workflow knowledge and understanding	Demonstrates deep technical and practical understanding of agentic AI concepts, workflows, and methods; clearly explains how AI actions and tool usage support tasks.	Shows good understanding of agentic AI workflows; provides logical reasoning for AI actions and task execution.	Demonstrates basic understanding of agentic AI concepts; workflow explanation is partial or incomplete.	Limited understanding of agentic AI workflows; explanations are unclear or inconsistent.	Fails to demonstrate understanding of agentic AI or workflow reasoning.	10	
Business Understanding and Relavance	Clearly identifies how agentic AI adds value to the chosen company and industry; integrates business insights into workflow solution; report and presentation are well-documented with detailed explanations.	Identifies AI value for the company; provides reasonable business context; documentation and explanation are good.	Provides some discussion of business relevance; integration of AI and business value is partial; documentation is basic.	Limited explanation of business relevance; workflow solution is weakly tied to business value; documentation is minimal.	Fails to demonstrate understanding of business value; workflow solution irrelevant or missing; documentation insufficient.	10	
					TOTAL	20	

Component Title	Agentic AI and Workflow Prototype				Percentage (%)	30	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (15-13)	Good (12-10)	Average (9-7)	Need Improvement (6-4)	Poor (3-0)		
Content Delivery and Q&A	Project content delivered clearly with strong insights on agentic AI workflows and	Project delivered clearly with solid insights on agentic AI and workflows; students answer	Project delivered clearly with moderate insights; students	Project delivered with weak insights; students struggle to	Project unclear, minimal insights; students	15	

	business applications; students answer all questions from lecturer and peers confidently.	**most questions effectively.**	**answer some questions.**	**answer questions.**	**cannot answer questions.**		
Report & Implementation	**Comprehensive report and fully functional implementation demonstrating agentic AI workflow; clear documentation of AI actions and business relevance.**	**Good report and working implementation; documentation mostly clear and relevant to workflow.**	**Basic report and partial implementation; documentation adequate but limited in workflow explanation.**	**Limited report or incomplete implementation; documentation weak or unclear.**	**Report missing or implementation non-functional; documentation absent or unclear.**	**15**	
					TOTAL	**30**	

Note to students: Please include the marking rubric when submitting your coursework.